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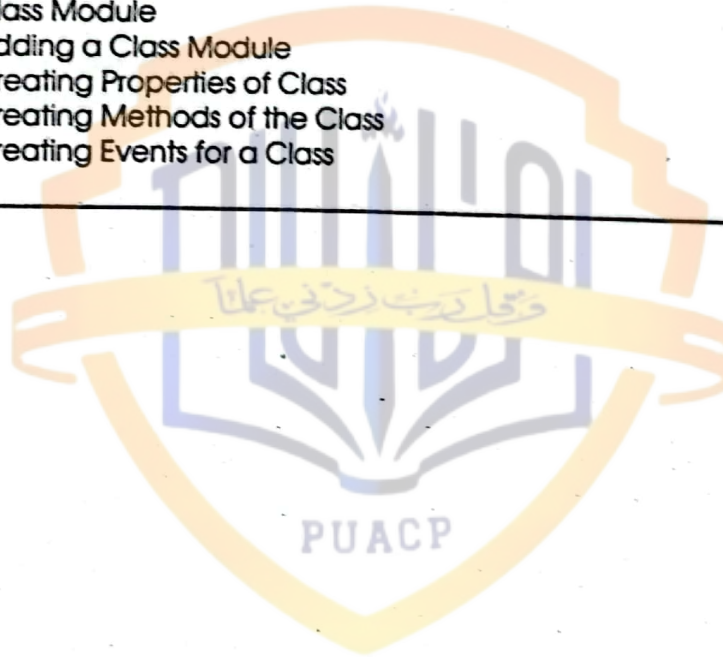
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18.1 Object Oriented Programming

Object oriented programming is a programming technique that follows real-world concepts. Programs are written on the basis of objects. An object is a set of data and functions. Object may represent a person, thing or place in real world. The data of an object describes the object and functions are the actions that can be performed on data. OOP allows programmers to create reusable objects that interact with each other to form complex entities.

18.2 Classes

Classes are the basis of all objects. A class is a collection of code that forms a design to create objects in an application. The code written in a class describes attributes and functions of objects that are created from class. Following are three major principles to create objects:

- Encapsulation
- Polymorphism
- Inheritance

18.2.1 Encapsulation

The information of an object such as internal data and code is hidden. Encapsulation isolates the internal complexity of an object from remaining application. For example, when a user sets Caption property of a form, he does not need to know how the string is stored.

18.2.2 Polymorphism

Polymorphism indicates the ability of an object to have methods with the same name but different content. For example, two command buttons on a form might both have a Paint method. The code in each object is different although the name of method is the same.

18.2.3 Inheritance

Inheritance is the ability of a subclass to use the characteristics of the parent class from which it is derived. If the characteristics of parent class change, subclass also inherits these characteristics. For example, if a new property IsBlank is added to a control, any subclasses based on the control will also have this property.

18.3 Creating a Class

Following are the basic steps to create a class in Visual Basic:

1. Add a class module and set its properties
2. Create the properties of the class
3. Create the methods of the class
4. Create events of the class

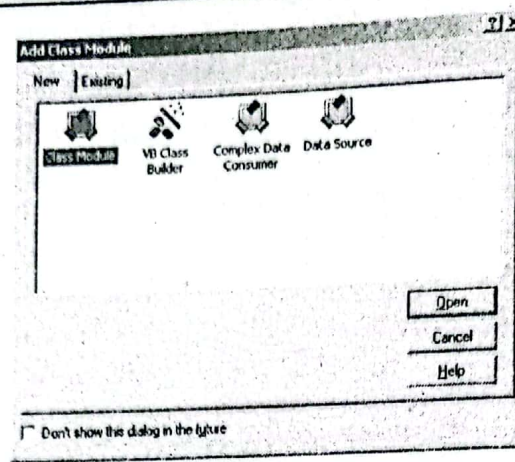
18.3.1 Class Module

Class modules is the base of object-oriented programming in Visual Basic. A class module contains the definition of a class. It includes property and method definitions. A class is created in a file with .CLS extension. The extension indicates that code is a class module. A class module and a standard code module are similar because both contain functionality that can be used by other modules within an application.

18.3.2 Adding a Class Module

A class module is added in the project to create a class. The following steps are performed to add a class module:

- Select Add Class Module from Projects menu. A dialog box will appear.



- Select Class Module option and click Open. A class module will be added.

18.3.3 Creating Properties of Class

Properties are used to pass data between a program and a class. Properties in a class module can be created in two ways:

Creating a Public Variable

Public variables are created by using **Public** keyword in **Declarations** section of class module. The following lines create two public variables:

```
Public AccNum As Integer
Public Balance As Integer
```

It is a simple method of creating property of a class. But it provides open access to the information in the class. It also does not enable the user to create read-only properties that are often required by a class.

A variable can also be declared as **Private**. The private variable can only be accessed by code in the class module.

```
Private AccNum As Integer
Private Balance As Integer
```

Creating a Property Procedure

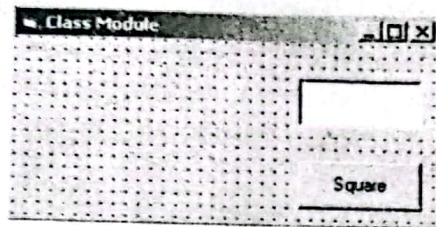
The recommended method of creating class property is to create property procedures that enforce data hiding. Data hiding is the ability to protect a part of an object's data and exposing the rest of data.

Visual Basic can create three types of property procedures:

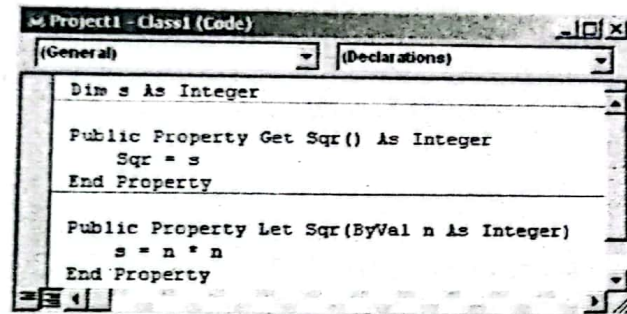
- **Property Let:** It is used when the value passed to the procedure itself is an object. It is the only procedure that cannot be created with Procedure dialog box.
- **Property Set:** It specifies an argument that contains the value passed to the property. The argument is passed by value to prevent the procedure from changing the value of the variable passed to the procedure.
- **Property Get:** It is used to return a value from class.

Practical 18.1

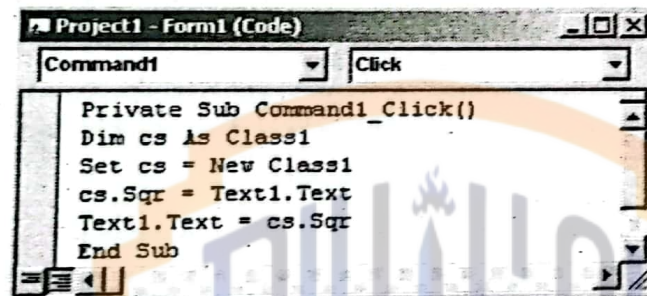
1. Start a new Standard EXE project.
2. Place a textbox and a command button on the form.
3. Design the form as follows:



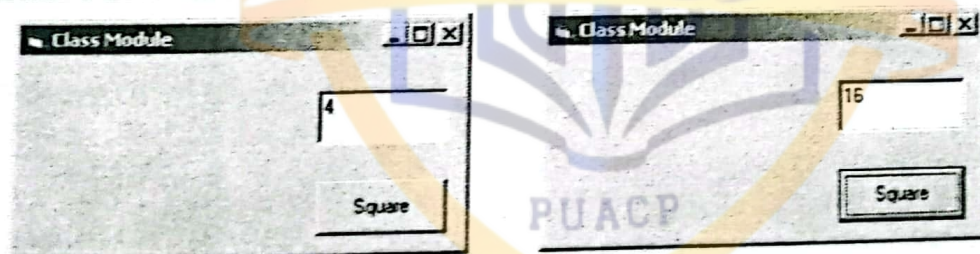
4. Select Project > Class Module. A class module will be added in the project.
5. Add the following code in class module:



6. Add the following code in Click event of the button on form:



7. Run the project.
8. Enter 4 in textbox and click button. The result will appear as follows:



How it Works?

The variable *s* is used to store the result. The Property Let procedure calculates the square of the number and assigns this value to *s* variable. Property Get procedure is used to retrieve the value from the class.

In code window of Form, following statement creates an object variable *cs* of *Class1*.
`Set cs = New Class1`

The following lines pass the number in text box to the class and retrieve it back from the class with the following statements:

```

cs.Sqr = Text1.Text
Text1.Text = cs.Sqr

```

The Property Set procedure takes an object as argument. The Set keyword assigns the object to internal object in class.

18.3.4 Creating Methods of the Class

After we create properties for a class we need to provide functionality to the class by adding methods to the class. We use the Public Sub and the Function procedures to create the methods of the class.

To create Public Sub or Function procedure we can use Procedure dialog box or declare the procedure by typing in code window of class module. For example, the following code can be used to create a method Name in class module that prompts user to enter valid name.

Example

```
Public Function Name (ByVal newstr As String)
    If IsNumeric(newstr) = True Or newstr = Empty Then
        MsgBox "Enter a valid name"
    End If
End Function
```

We can create private procedures by specifying the keyword Private in the procedure declaration. This procedure can be used inside the class but is unavailable to the rest of the program.

18.3.5 Creating Events for a Class

The classes in Visual Basic can also have events. Events tell the application about the occurrence of some action in the class. Two important events are as follows:

Initialize

Initialize event occurs when an instance of a class is created. It is triggered before any properties are set. It is used to initialize any data of the class. It can also be used to load forms used by the class.

Syntax: Private Sub Object_Initialize()

Example

The following example in Initialize event of a class creates and opens a connection and a recordset.

```
Private Sub Class_Initialize()
    Set Conn = New ADODB.Connection
    Conn.Open "Provider=Microsoft.Jet.OLEDB.3.51;" & _
        "Data Source=C:\Program Files\Microsoft Visual Studio\VB98\Nwind.mdb"
    Set rec = New ADODB.Recordset
    rec.Open "Customers", Conn
End Sub
```

Terminate

Terminate event occurs when the object is set to 'Nothing' or when the last reference to the object goes out of scope. It is used to unload forms or perform tasks that should occur when the class terminates.

Syntax: Private Sub Object_Terminate()

Example

```
Private Sub Class_Terminate()
    Unload Form1
End Sub
```